

UTM CSC343: Introduction to Databases
Midterm Examination October 28, 2004

Student Number: _____

Last Name: _____ First Name: _____

This exam consists of 5 double sided pages. Unless otherwise specified, assume that you are using a SQL standard system. You are allowed one $8\frac{1}{2} \times 11$ double sided aid sheet. **You might want to remove the last sheet.**

1. (5 Marks: Medium) Consider the **Beers**(**bar**, **beer**, **price**) relation and the following expression from lecture:

$\sigma_{beer1 \neq beer}((\rho_S(bar, beer1, price)(Beers)) \bowtie Beers)$

- (a) What is the schema for this expression (ie. what fields are returned).

- (b) Write an equivalent expression using only $\rho, \pi, \sigma, \times$. Full marks only for the same schema and resulting rows.

2. (6 Marks: Easy) Write down the results of the execution of the following:
If you do not understand HAVING, then (for 1/2 marks) cross it off and solve the simpler problem.

```
DROP TABLE q;  
CREATE TABLE q ( r1 real, r2 real);
```

```
INSERT INTO q VALUES(-1,-2);  
INSERT INTO q VALUES(1,2);  
INSERT INTO q VALUES(1,2);  
INSERT INTO q VALUES(2,4);  
INSERT INTO q VALUES(2,4);  
INSERT INTO q VALUES(3,6);  
INSERT INTO q VALUES(3,6);  
INSERT INTO q VALUES(3,1);  
INSERT INTO q VALUES(3,-2);  
INSERT INTO q VALUES(3,6);  
INSERT INTO q VALUES(3,8);  
INSERT INTO q VALUES(4,1);  
INSERT INTO q VALUES(4,0);  
INSERT INTO q VALUES(4,-7);  
INSERT INTO q VALUES(5,15);  
INSERT INTO q VALUES(5,10);  
INSERT INTO q VALUES(12,-11);  
INSERT INTO q VALUES(12,1);  
INSERT INTO q VALUES(12,-7);  
INSERT INTO q VALUES(13,1);  
INSERT INTO q VALUES(13,3);
```

```
SELECT r1, SUM(r2)  
FROM q  
WHERE r1>2  
GROUP BY r1  
HAVING MAX(r2)>=3;
```

For the next questions we will be considering the Flight Database from assignment 1. See the last page of the midterm for it's description.

3. (10 Marks: Easy) Write the ER diagram which gave rise to the Flight Database schema on the last page of the midterm.

4. (10 Marks: Mixed) Complete the SQL below. Include all of the following constraints. Use any reasonable assumption about text field sizes and use appropriate SQL data types. **Hint:** I have included some notes at the end of this exam for you.

- (a) Primary keys as described above.
- (b) Obvious foreign key constraints, as well, if a flight is deleted, then so are all the bookings for the flight. If a customer is removed from the database, then so is all their bookings. If a customer's sin is changed, then those changes should appear in the bookings table.
- (c) Departure Time is always before arrival time.
- (d) The capacity of an aircraft is always at least 0.
- (e) The row is a number, the seat is a single character (ie 23,F), neither can be null.
- (f) No two passengers can be booked in the same (row,seat) pair for the same flight.

```
CREATE TABLE booking (
```

```
    sin
```

```
    flightNum
```

```
    day
```

```
    row
```

```
    seat
```

```
);
```

```

CREATE TABLE aircraft (
    acName
    capacity

);
CREATE TABLE flight (
    flightNum
    day
    acName
    leavesFrom
    arrivesAt
    departureTime
    arrivalTime

);
CREATE TABLE customer (
    sin
    name
    address
    phone

);

```

5. (10 Marks: Hard) Consider the following query

```

SELECT C.name, C.address
FROM Customer C
WHERE (SELECT COUNT(DISTINCT F.arrivesAt)
      FROM Flight F NATURAL JOIN Booking B
      WHERE C.sin=B.sin)>4;

```

- (a) Describe simply (and in english) what this query returns. No marks for things like: *The rows in the customer table for which 4 is less than the count of the...*
- (b) Write an equivalent SQL query which does not use a sub-select (ie only a single select query). Your query must return the same rows and have the same schema.

6. (30 Marks: Mixed) Write RA and SQL for each of the following

(a) (Easy) Find all customers (sin, name) on flight 1158, October 28, 2004.

(b) (Medium) Find all customers (sin, name, phone, flightNum) that are leaving from or arriving at Toronto's Pearson airport on October 28, 2004.

(c) (Hard) Find customers that have never flown on a B737 (acName='B737').

Hint: A simpler question is 'Find sin's of customers that have flown a B737'.

The following is the description of the Flight Database

booking (sin, flightNum, day, row, seat)
flight(flightNum,day, acName, leavesFrom, arrivesAt, departureTime, arrivalTime)

customer(sin, name, address, phone)

aircraft(acName, capacity) Each Flight has an aircraft assigned to it.

The following is an excerpt from the course notes.

```
CREATE TABLE course (
    code CHAR(6) PRIMARY KEY,
    title VARCHAR(20),
    enrollLimit INTEGER
        DEFAULT 0
    CONSTRAINT enrollCons CHECK(enrollLimit>=0)
);
CREATE TABLE student (
    sid CHAR(9),
    name CHAR(20),
    account FLOAT
        DEFAULT 0.0
    CHECK(account>=0),
    PRIMARY KEY(sid)
);
CREATE TABLE class (
    sid CHAR(9) REFERENCES student(sid)
        ON DELETE CASCADE
        ON UPDATE CASCADE,
    code CHAR(6) REFERENCES course(code)
        ON UPDATE CASCADE,
    mark INT
        DEFAULT 0.0
        NOT NULL
    CHECK(mark>=0 AND mark<=100),
    PRIMARY KEY(sid,code)
);
```